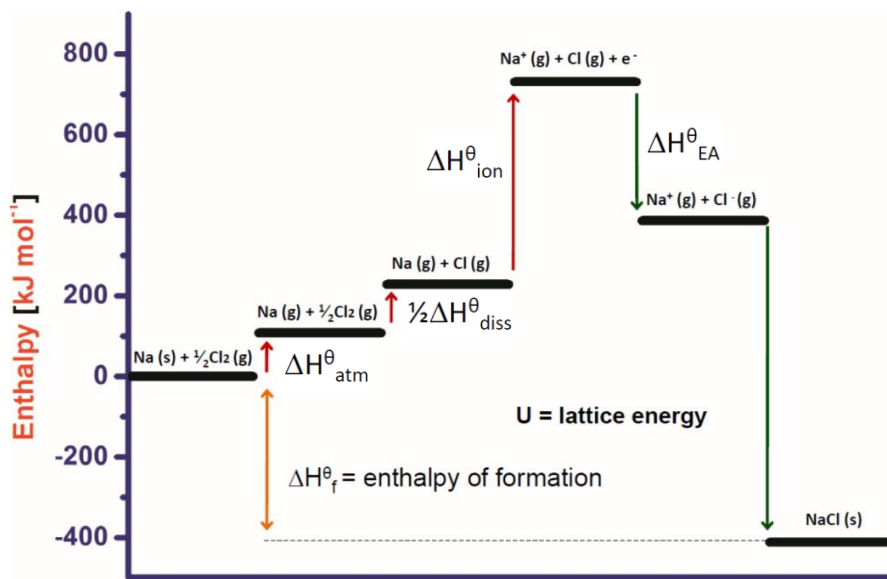


- Which forces have to be taken into account when we calculate the lattice energy of an ionic solid ?
 - Coulombic attractive and repulsive energies, which account about the 90% of the total lattice energy;
 - Additional repulsive energy resulting from repulsion between the overlapping outer electron density of adjacent ions;
 - Other minor contributions as van der Waals energy.
- Draw the Born-Haber cycle for NaCl.



- What is the Madelung constant? Can we apply it at different crystals?

The Madelung constant is used in the Born-Landé equation to take into account the different arrangement of cations and anions in a crystal lattice. The Madelung constant is a purely geometrical constant and therefore crystals with the same arrangement of cations and ions have the same Madelung constant.
- Which are the possible geometries for substances with coordination number 8.

The most common arrangement of complexes with coordination number 8 is the cubic geometry

5) By taking in consideration the position of the elements in the periodic table, Indicate with a cross the true sentences :

- ☐ Lithium has higher melting point than beryllium.
- ☐ Mg has larger atomic radius than Na
- ☒ The first ionization energy is higher for Mg than for Na
- ☐ The second ionization energy is higher for Mg than for Na
- ☐ The first ionization energy is higher for Cs than for Na

6) Explain the cohesive energy of metals

Metallic substances consist of atoms with low electronegativity. These atoms can share their valence electrons with all the material. The electrons can move freely within the material so that metals typically show good electron conductivity. Metallic material are held together by the strong forces of attraction between the positive nuclei and the delocalized electrons.

7) What are the most important properties for a solvent?

- a) The temperature-pressure range over which it is a liquid.
- b) Its dielectric constant.
- c) Its donor and acceptor properties: solvent polarity
- d) Its protonic (Brønsted) acidity or basicity.
- e) The nature and extent of autodissociation

8) Taken the reaction

... $\text{H}_3\text{PO}_4 = \text{H}^+ + \text{H}_2\text{PO}_4^-$, is H_3PO_4 a base or an acid? Why?

H_3PO_4 is a proton donor, thus is an acid

(Brønsted definition).

... $\text{CaO} + \text{SiO}_2 = \text{CaSiO}_3$, is SiO_2 a base or an acid? Why?

CaO is a base, because it can donate an oxygen atom to silicon

(Lux-Flood definition).

... $\text{PF}_5 + 2\text{HF} (\text{solvent}) = \text{PF}_6^- + \text{H}_2\text{F}^+$, is PF_5 a base or an acid? Why?

PF_5 increases the concentration of the solvent cationic species, thus it is an acid according to the solvent system definition.

... $\text{AlCl}_3 + \text{PF}_3 = \text{Cl}_3\text{Al}:\text{PF}_3$, is PF_3 a base or an acid? Why?

P uses its lone pair to form a P-Al bond, thus PF_3 is a base

(Lewis definition).

... $3\text{Na}_2\text{O} + \text{P}_2\text{O}_5 = 2 \text{Na}_3\text{PO}_4$, is Na_2O a base or an acid? Why?

Na_2O is a base, because it can donate an oxygen atom to silicon

(Lux-Flood definition).

... $10 \text{S} + 4 \text{NH}_3 (\text{solvent}) = \text{S}_6^{2-} + \text{S}_4\text{N}^- + 3 \text{NH}_4^+$, is S a base or an acid? Why?

S increases the concentration of the solvent cationic species, thus it is an acid according to the solvent system definition